

DeepFER: Facial Emotion Recognition Using Deep Learning

1. Project Name

DeepFER: Facial Emotion Recognition Using Deep Learning

2. Objective

To develop a deep learning-based system that classifies facial expressions into **7 emotions: Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise** using CNN and transfer learning techniques.

3. My Understanding

Facial Emotion Recognition is a computer vision problem that identifies emotions from facial expressions. The project aims to build a model capable of learning facial patterns and predicting emotions while handling challenges such as **class imbalance, low-resolution images, and visually similar expressions**.

4. Dataset

The dataset contains **35,887 facial images** across 7 emotion classes.

Property	Details
Images	35,887
Classes	7
Image Size	48 × 48
Image Type	Grayscale
Training	25,942

Validation **2,879**

Test **7,066**

The dataset was found to be imbalanced, with **Happy** having the highest number of samples and **Disgust** the lowest.

5. Project Workflow

Dataset → Data Integrity Check → EDA → Preprocessing → Augmentation → Class Weighting → Custom CNN → Transfer Learning → Training & Optimization → Evaluation → Error Analysis → Prediction Demo → Real-Time Application Design → Conclusion

6. Approach

Data Preparation

- Checked image integrity and dimensions.
- Analyzed class distribution and sample images.
- Confirmed **35,887 valid images**.

Preprocessing

- **48×48 grayscale** images
- Pixel normalization to **0–1**
- One-hot encoded labels
- Data augmentation using rotation, shifting, zoom, and flipping
- Class weighting to address imbalance

Model Development

Two approaches were evaluated:

1. Custom CNN

Three convolution blocks with Batch Normalization, MaxPooling, Dropout, and Dense layers.

2. MobileNetV2 Transfer Learning

ImageNet-pretrained MobileNetV2 with a custom classification head.

Optimization

Used **Adam**, **Early Stopping**, **ReduceLROnPlateau**, **Model Checkpointing**, **Dropout**, and **Batch Normalization**.

7. Key Results

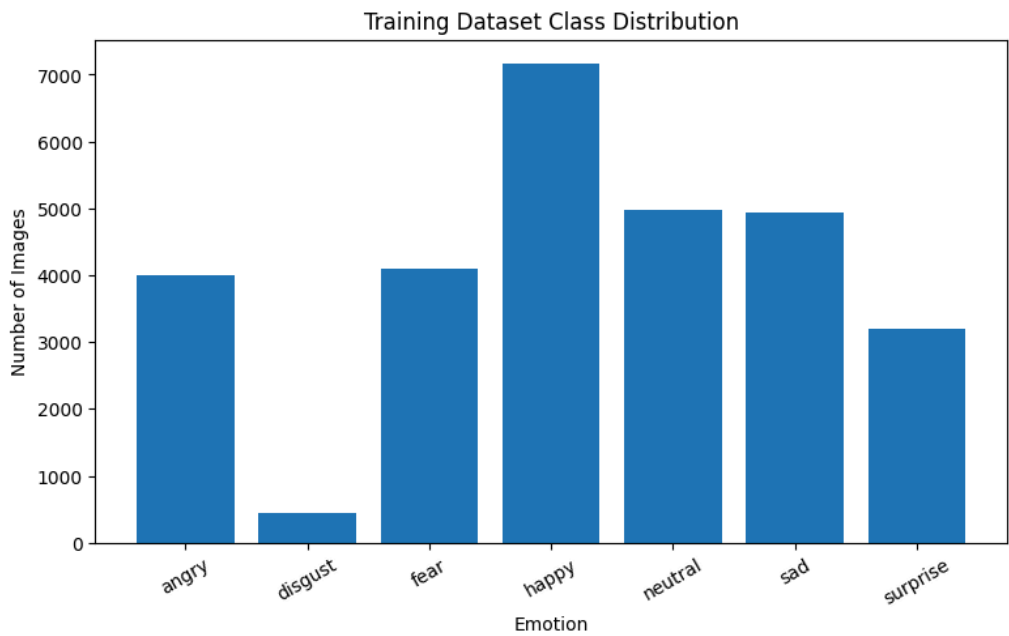
Model	Test Accuracy	Macro F1
Custom CNN	50.38%	42.99%
MobileNetV2 (Frozen)	40.70%	34.45%

The **Custom CNN** performed **better** than the frozen MobileNetV2 for this dataset.

8. Key Screenshots

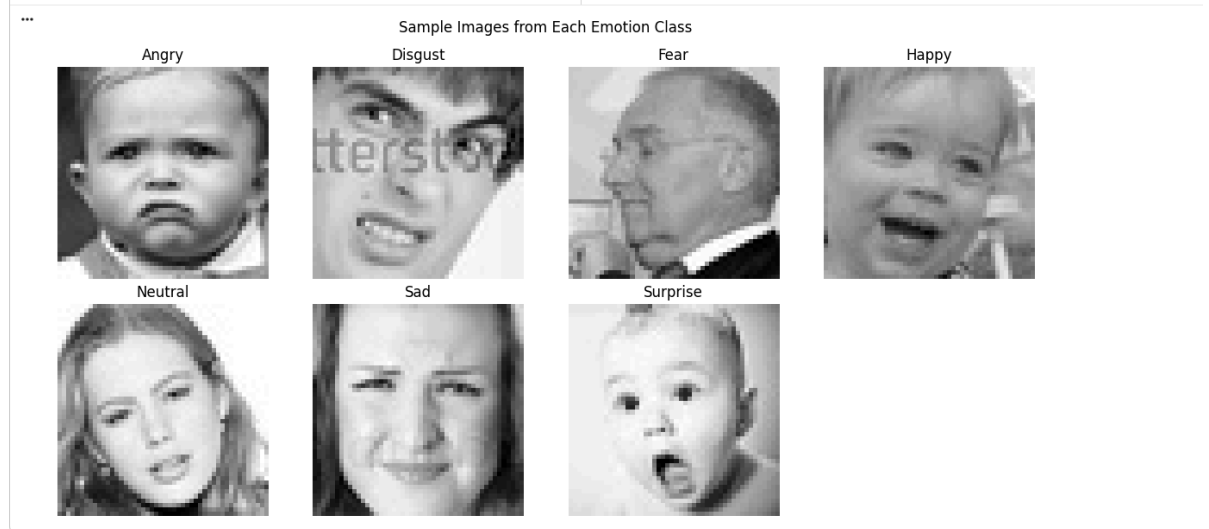
1. Dataset Class Distribution

Emotion Classes: ['angry', 'disgust', 'fear', 'happy', 'neutral', 'sad', 'surprise']
Total Training Images: 28821



2. Sample Images from All 7 Classes

```
plt.tight_layout()  
plt.show()
```



Insight

The sample images show noticeable variation in facial appearance and expressions across all seven classes. The images are grayscale

3. CNN Architecture / Model Summary

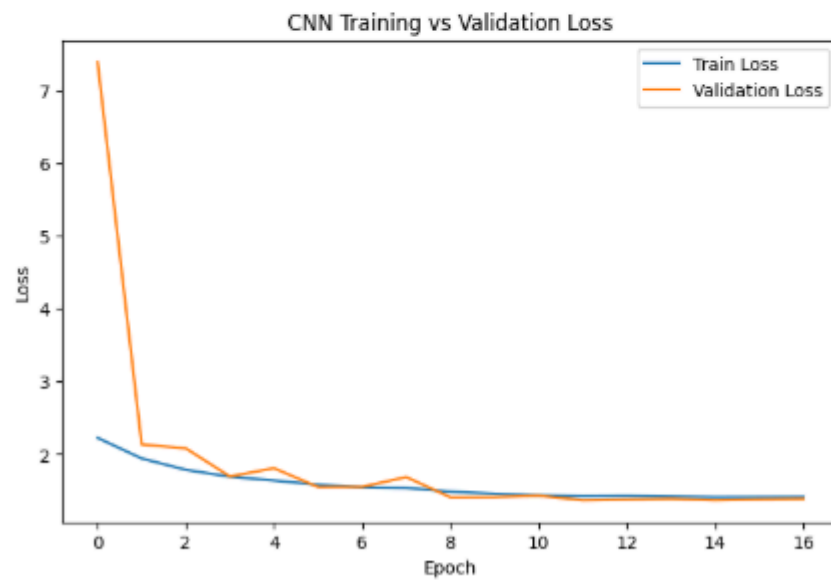
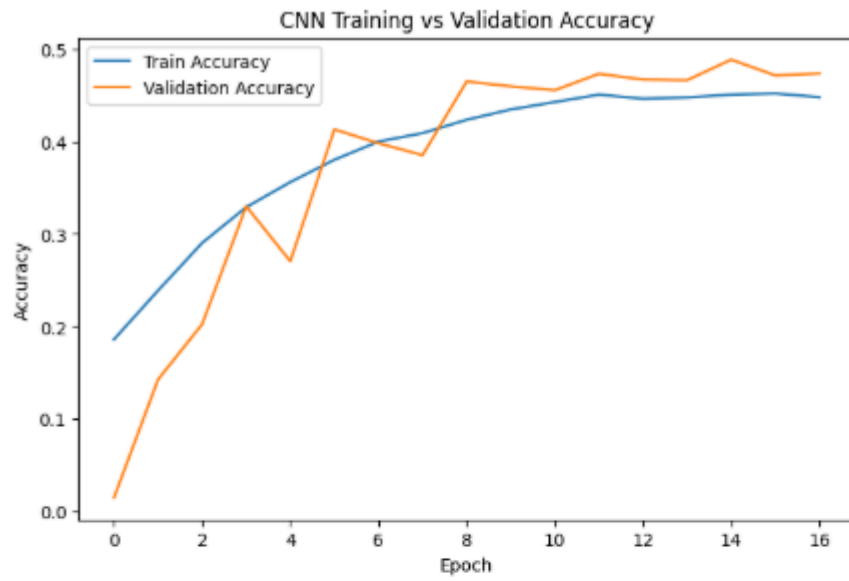
Model: "sequential"

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Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 48, 48, 32)	320
batch_normalization (BatchNormalization)	(None, 48, 48, 32)	128
max_pooling2d (MaxPooling2D)	(None, 24, 24, 32)	0
dropout (Dropout)	(None, 24, 24, 32)	0
conv2d_1 (Conv2D)	(None, 24, 24, 64)	18,496
batch_normalization_1 (BatchNormalization)	(None, 24, 24, 64)	256
max_pooling2d_1 (MaxPooling2D)	(None, 12, 12, 64)	0
dropout_1 (Dropout)	(None, 12, 12, 64)	0
conv2d_2 (Conv2D)	(None, 12, 12, 128)	73,856
batch_normalization_2 (BatchNormalization)	(None, 12, 12, 128)	512
max_pooling2d_2 (MaxPooling2D)	(None, 6, 6, 128)	0
dropout_2 (Dropout)	(None, 6, 6, 128)	0
flatten (Flatten)	(None, 4608)	0
dense (Dense)	(None, 128)	589,952
batch_normalization_3 (BatchNormalization)	(None, 128)	512
dropout_3 (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 7)	903

Total params: 684,935 (2.61 MB)
Trainable params: 684,231 (2.61 MB)
Non-trainable params: 704 (2.75 KB)

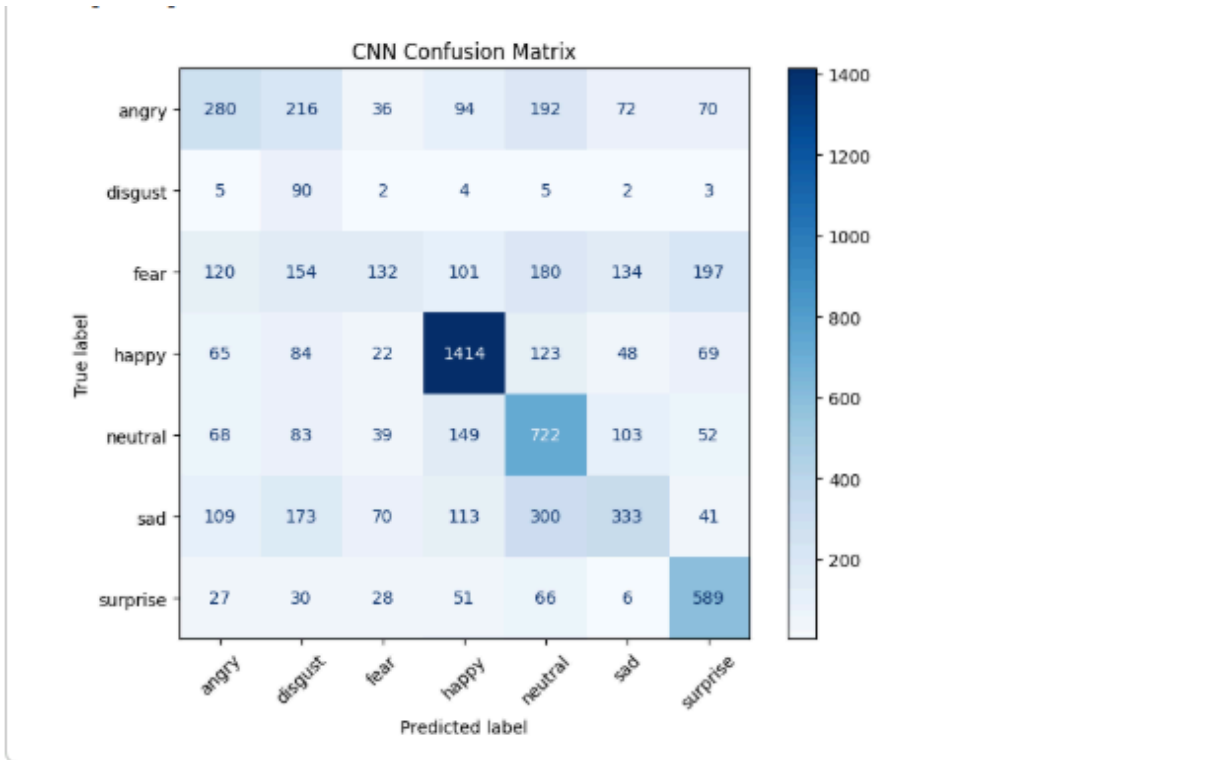
4. Training Accuracy & Loss Curves



Insight

The CNN shows **steady learning with decreasing loss**, while validation accuracy reaches about **49%** before Early Stopping.

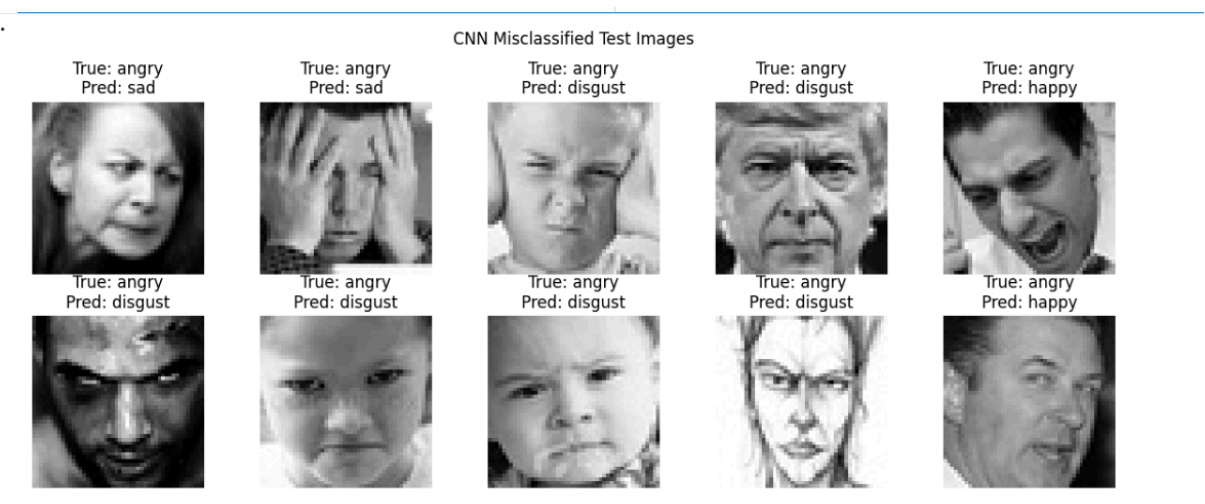
5. CNN Confusion Matrix



Insight

The baseline CNN achieved **50.38% test accuracy**. Performance is stronger for **Happy**, while minority and visually similar emotions remain challenging.

6. Prediction / Error Analysis Demo



Error Analysis Insight

10 misclassified samples show strong overlap between **Angry, Sad, and Disgust**, indicating that subtle facial expressions are difficult to distinguish in low-resolution grayscale images.

9. Problems Faced

- **Class imbalance**, particularly the small number of Disgust samples.
 - **48×48 low-resolution images** made subtle expressions difficult to distinguish.
 - Significant confusion between **Angry, Disgust, Fear, and Sad**.
 - Transfer learning with frozen MobileNetV2 performed worse than the custom CNN.
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10. Learnings

- Learned how CNNs automatically extract **hierarchical visual features** from images.
 - Understood the importance of **augmentation and class weighting** for imbalanced image datasets.
 - Learned how **Early Stopping and learning-rate scheduling** improve training efficiency.
 - Learned that pretrained models do not always outperform a **task-specific CNN**, especially when the source and target image characteristics differ.
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11. Future Improvements

- Use **higher-resolution facial images**.
 - Improve class balancing and minority-class representation.
 - Experiment with stronger CNN architectures.
 - Fine-tune pretrained models more extensively.
 - Develop and deploy a complete **real-time emotion recognition application**.
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12. Conclusion

DeepFER successfully developed a **7-class facial emotion recognition system** using deep learning. The Custom CNN achieved **50.38% test accuracy and 42.99% macro F1**, outperforming the frozen MobileNetV2 approach.

The project demonstrates a complete workflow from **data preparation and preprocessing to model development, evaluation, error analysis, and application design**, while also highlighting the challenges of emotion recognition from low-resolution and imbalanced data.